University of Essex

Department of Mathematical Sciences

MA981: DISSERTATION

Forecasting Post-Meal Glucose Spikes Across Multiple Time Windows using Multimodal Health Data and AI

Pulak Saha
2411750

Supervisor: Dr Saideh Ferdowsi

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Abstract

The postprandial glycaemic response (PPGR), which is the increase in blood glucose after eating, is a key sign of metabolic health and an early sign of moving towards prediabetes and type 2 diabetes. People often have different PPGR values even when they eat the same meal. Same food, influenced by lifestyle, physiology, and metabolic state. This variability limits the effectiveness of routine dietary advice and indicates the need for personalised and data driven prediction tools.

The purpose of this dissertation is to determine the potential for predicting glucose spike events by analyzing real-time and non-invasive sources of information through On-going Glucose Measurement data, Meal Level Features, clinical biomarkers, Activity Measure, and Image Based Food Embedding collected within the CGMacros dataset. A complete machine-learning approach to modeling predictions has developed a pipeline with Multimodal Feature Engineering, Participant Aware Data Splits, and Algorithms to compare models using XGBoost, LightGBM, CatBoost, Long Short-Term Memory (LSTM), and Transformers. We produced glucose spike predictions at 1, 3, and 6 hours following consumption of the meal.

The Transformer achieved the strongest performance, with test set R squared values of 0.6727, 0.6645, and 0.6486 for the three prediction horizons. These results show accurate short term prediction and stable performance over longer intervals. Recent CGM dynamics were the most influential predictors, while nutritional and clinical measures contributed additional value.

Overall, the findings demonstrate that personalised glucose spike prediction is feasible using simple and scalable data sources, without reliance on costly tests. This supports the development of practical personalised nutrition tools for real world health applications.

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CHAPTER 1

Introduction

1.1 Background and Context

Metabolic diseases such as Type 2 Diabetes (T2D), obesity, and metabolic syndrome represent major and growing public health challenges worldwide. Modern lifestyle patterns, including increased consumption of processed foods and reduced levels of physical activity, have contributed to rising rates of impaired glucose regulation [1, 2]. More than 537 million adults currently live with diabetes, and this number is projected to reach 640 million by 2030 [3]. These conditions are associated with serious complications including cardiovascular disease, neuropathy, kidney failure, and reduced quality of life [4].

Metabolic disorders are characterized by diminished metabolism of blood glucose after eating and an inability to regulate blood glucose levels with foods eaten after consumption of the foods. Healthy individuals will use hormones, mostly insulin, along with other particular hormones to keep the blood glucose levels, approximately equal between blood glucose levels and energy requirements of the body. When this system fails to function or become damaged, the blood glucose levels can rapidly increase or remain abnormally elevated following meals. The increases in blood glucose levels during or following a meal are directly related to systemic inflammation, oxidative stress, insulin insensitivity, and long-term metabolic failure [5].

Traditional clinical markers such as fasting glucose and HbA1c provide useful summaries of metabolic health but do not capture short-term fluctuations in glucose dynamics. Con-

tinuous Glucose Monitoring (CGM) has transformed the study of glycaemic regulation by providing high-frequency, real-world measurements of glucose levels throughout the day [6]. CGM enables detailed examination of postprandial glycaemic response (PPGR), which is considered a sensitive and informative early indicator of cardiometabolic risk.

A key outcome from recent research is that individuals show highly variable PPGRs even when eating identical meals [7, 8]. These variations arise from multiple factors including nutritional composition, biological timing, physical activity, sleep, physique, and insulin sensitivity. As a result, general dietary recommendations often lack precision and do not reflect specific glycaemic physiology.

In cases of Metabolic disorder, the regulation of blood glucose becomes impaired, resulting in an inability to clear glucose from the circulatory system following meal consumption. In September 2019, researchers in Germany published an article in the Journal of Personalised Medicine that identified Machine Learning as a means to combine many Multimodal Data Sources (CGM, Dietary Logs, Clinical Biomarkers and Smart Devices) into one Personalised Nutrition model. The study suggested that using Machine Learning as a method to identify Complex and Non-Linear Relationships among these Multimodal Data Sources provides a greater predictive ability in relation to a person's Postprandial Glucose Levels (PPG), which is evidenced by the fact that Microbiome and Genome Data were included in many studies researching Machine Learning-based nutritional models, however, they were too expensive to be used on a regular basis and not widely available to be implemented in a scalable manner. To develop scalable and effective Personalised Nutrition Models, researchers need to determine what Simple and Widely Available Data Features can accurately predict Relative Glucose Delta Spikes following meals.

This Dissertation investigates how accurately a Prediction Model can predict the onset and intensity of Glucose Delta Spikes through the use of Multimodal Data Sources (CGMacros) and correlates the Results with Clinical Significance.

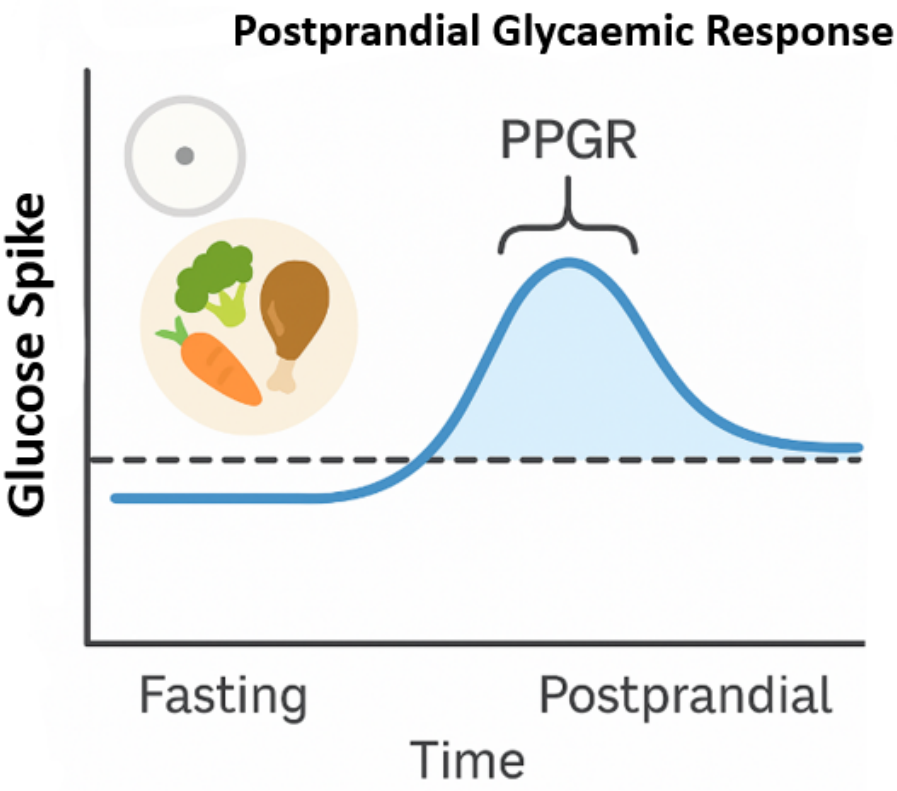

Figure 1.1: Postprandial glycaemic response (PPGR): rise in blood glucose following food intake.

1.2 The Specific Problem

Although CGM technology provides detailed insight into glucose dynamics, there remains a need to improve the ability to predict how an individual's glucose level will change after a meal. Current clinical and nutritional guidance remains largely generalised even though individuals differ substantially in their responses to food.

To advance personalised nutrition, it is necessary to develop machine learning models that can predict glucose changes across multiple time horizons and identify the factors that most strongly influence glycaemic outcomes. This requires modelling glucose deltas at 1 hour, 3 hours, and 6 hours after meal consumption to capture fast, medium, and delayed responses.

1.3 The Knowledge Gap

Two central gaps motivate this dissertation.

1.3.1 Scalable Glucose Spike Prediction

There have been numerous studies conducted on the various types of data used to build predictive models about the development of the glucose spike, including microbiomes, metabolomics and laboratory testing data. While many of these predictive models have been shown to work well, they require large data sets that are not readily available in a clinical or consumer setting. Therefore, it is important to determine how well glucose spikes can be predicted using inexpensive, user-friendly, and easily accessible data features, such as the content of one or more meals, glucose monitoring history and standard clinical indicators, to determine how accurately glucose levels can be predicted.

1.3.2 Understanding Feature Drivers of PPGR

Based on previous research into carbohydrate, fiber, sleep activity, and anthropometric variables affecting blood glucose responses, it has been suggested that all five variables are significant; however, the degree of their influence is greatly varied. A more thorough and comprehensive exploration of these factors will be needed to identify common or 'predictable' PPGR-associated characteristics, and to understand the differences in their effects over time.

1.4 Aim and Objectives

In this research paper, we present our work in developing and validating machine learning models that can predict short-term increases in blood glucose levels following the consumption of meals by utilizing the exact meal level data from a variety of sources. Initially, we build an extensive Preprocessing Pipeline that draws on CGM-Based Time Series Data, Food Logs, as well as Personal Profiles of each participant before concentrating on predicting changes to blood glucose levels occurring 1, 3 and 6 hours post consumption. We train and assess three popular Gradient Boosting Models (GBM), namely XGBoost, LightGBM and CatBoost, and two Deep Learning Models, namely LSTM and Transformer, so as to evaluate the

performance capabilities between each one. All models are validated by using Group Aware Validation Methodology to ensure that they are not trained by using shared information across multiple subjects. At the conclusion of this dissertation, an investigation is performed to assess which dietary, lifestyle and physiological variables influence Postprandial Glucose Response (PGR), using Feature Importance and SHAP Value Analysis methods.

1.5 Research Questions

- **RQ1:** How accurately can machine learning models predict glucose delta spikes at 1 hour, 3 hours, and 6 hours after eating?
- **RQ2:** Which features, including meal composition, glucose history, circadian context, and participant-level biomarkers, most strongly influence PPGR?
- **RQ3:** Which gradient boosting model or deep sequence learning model provides the highest predictive performance for glucose delta modelling?
- **RQ4:** How does predictive accuracy change across short, medium, and long postprandial time horizons?

1.6 Contributions

This dissertation presents several significant contributions. This work presents a scalable prediction pipeline that utilises readily available data sources and eliminates the need for microbiome or genomic sequencing. This study illustrates a robust multimodal modelling approach that integrates tabular features with embeddings derived from meal images. In addition, the dissertation provides a thorough comparison of leading gradient boosting algorithms for predicting glucose delta values. This study identifies the nutritional and physiological variables that significantly affect postprandial glucose responses across various prediction time frames. Finally, it discusses the broader implications of these findings for personalised nutrition, dietary decision support, and real-time glucose forecasting applications.

CHAPTER 2

Literature Review

In the fields of metabolic data science, personalised nutrition, and digital health, postprandial glucose response prediction has grown in importance. The development of Continuous Glucose Monitoring (CGM) devices has made it possible to measure glucose changes in real time with great resolution. This has opened up new possibilities for the development of machine learning models that can forecast abrupt changes in blood sugar levels, which are frequently quantified as short-term postprandial glucose spikes or glucose delta.

A consistent theme across the literature is the use of multimodal data, including meal composition, glucose history, lifestyle behaviours, and sensor-based features, to model individual differences in post-meal glucose regulation. The following sections summarise key findings from recent research focused specifically on predicting glucose spikes.

Machine Learning for Postprandial Glucose Prediction

Zeevi et al. [7] conducted one of the most influential studies in personalised nutrition by demonstrating large individual variability in post-meal glucose responses, even among people consuming identical foods. Their model combined meal features, blood markers, anthropometrics, and microbiome data using a gradient boosting machine and achieved strong predictive performance. Although the microbiome was a major contributor in their framework, the study established that meal-level and clinical features alone contain substantial predictive value for glucose spikes.

Mendes-Soares et al. [8] extended this work in a U.S.-based cohort, confirming the high individual variability in postprandial glycaemic response (PPGR). Their models used gradient boosting methods with features derived from CGM signals, nutritional composition, and simple clinical traits. The study highlighted that carbohydrate content, glucose history, and meal timing were among the strongest predictors of glucose elevation after eating.

Carletti et al. [9] developed a multimodal machine learning model to distinguish individuals with normal, prediabetic, and diabetic glucose regulation. Using XGBoost with wearable sensor data, dietary logs, and electronic health records, their system achieved an AUC of 0.96 on internal validation. Although the primary outcome was classification rather than regression, their findings confirmed that combining behavioural and physiological features significantly improves the detection of abnormal glucose patterns.

A large-scale study published in *Frontiers in Nutrition* [10] focused on predicting meal-specific glucose responses in over 1,000 participants using CGM, smartphone food logs, and activity data. The researchers developed an XGBoost regression model that achieved a Pearson correlation of 0.71 between predicted and observed PPGR. Their feature importance analysis revealed that glycaemic history (pre-meal glucose patterns) and macronutrients (carbohydrates, fibre, protein) were the most influential predictors, supporting the value of time-series and nutritional features.

Burks et al. [11] introduced chronobiological features representing daily glucose rhythms and biological clocks derived from CGM data collected from more than 8,000 individuals with type 2 diabetes. Incorporating time-of-day patterns and rhythm metrics into an XGBoost model improved predictive accuracy for glucose fluctuations. Their work demonstrated that circadian timing plays a major role in glucose regulation and should be included in PPGR prediction models.

Kong et al. [12] proposed a double-layer simplified complex interval neural network stacking model for CGM-based glucose prediction. Their architecture processed interval-valued features to better handle uncertainty in sensor readings. The stacked model outperformed both traditional neural networks and single-layer interval models, showing that uncertainty-aware representations can enhance glucose spike prediction.

Lim et al. [13] introduced a deep learning approach that predicts near-term glucose changes using only lifestyle logs (meal intake, steps, and activity) without requiring real-time CGM values. Their dual-attention LSTM achieved a root mean squared error of 19.49 mg/dL,

demonstrating that meaningful glucose predictions can be generated from behavioural features alone. This supports the potential for smartphone-only personalised nutrition tools.

Rosales et al. [14] developed a hybrid CNN–LSTM architecture using CGM time-series windows to predict short-term glucose changes. Their model emphasised local temporal dynamics and achieved higher accuracy than traditional recurrent networks. This reinforces the importance of capturing rapid transitions in CGM data when modelling glucose spikes.

Recent studies continue to highlight the complexity and individuality of postprandial glucose responses. Bolle et al. [15] demonstrated that deep learning models trained on high-resolution CGM data can accurately forecast short-term glucose trajectories, outperforming traditional autoregressive methods. Their work emphasized the role of temporal features and physiological dynamics in shaping glucose excursions.

In a large-scale nutritional intervention, Berry et al. [16] collected CGM, meal logs, sleep, genetics, and activity profiles from over 1,000 participants. Using gradient boosting and mixed-effects models, they showed that individual-specific factors are often stronger determinants of PPGR than nutrient composition alone, reinforcing the need for personalized prediction.

Liang et al. [17] proposed a hybrid CNN-LSTM learning architecture capable of capturing both spatial nutrient patterns and temporal glycaemic evolution. Their model achieved state-of-the-art performance in glucose forecasting using CGM signals combined with lifestyle markers such as meals and exercise.

Sun et al. [18] explored the contribution of circadian rhythms to glycaemic variability. Using time-aware deep learning models, they found that incorporating time-of-day embeddings significantly improves prediction of glucose spikes, aligning with evidence that hormonal sensitivity changes across the day.

A comprehensive scoping review by Contreras and Vehi [19] highlighted the rapid growth of machine learning approaches for type 1 and type 2 diabetes glucose forecasting. They reported that models integrating exogenous inputs meals, insulin, activity consistently outperform sensor-only models, and they identified feature engineering as a major performance driver.

In another influential study, Atkinson et al. [20] investigated the reproducibility of personalized PPGR patterns and found that certain food responses are stable over time within individuals but highly variable between individuals. This supports the motivation for

models that use person-specific features rather than population averages.

Finally, Zhao et al. [21] developed an interpretable XGBoost-based PPGR model that integrated CGM features, macronutrient composition, and simple clinical covariates. Their system achieved high predictive accuracy while maintaining transparency through SHAP-based explanations, making it suitable for diet feedback applications and personalized nutrition.

Summary

The research unequivocally demonstrates that using contemporary machine learning algorithms to predict glucose spikes is possible, particularly when multimodal information are included. Depending on whether the input is mostly tabular or time-series data, both ensemble models and deep learning architectures perform well. Crucially, the majority of predictive power is derived from easily accessible and basic characteristics like time of day, glucose history, and macronutrient composition, indicating that scalable glucose spike prediction is possible without the need for costly biological data. The current dissertation, which focusses on developing machine learning models to predict glucose delta spikes (one, three, and six hours after meals) using multimodal characteristics collected from the CGMacros dataset, is based on this findings.

CHAPTER 3

Methodology and Data

The whole methodological approach for creating postprandial glucose response prediction models is described in this chapter. The pipeline is intended to limit data loss, represent real-world personalised nutrition scenarios, and guarantee that the models capture true physiological patterns rather than participant-specific characteristics. Every phase of the procedure has been designed to be transparent, repeatable, and scientifically sound, from the processing of raw data to the creation and assessment of models.

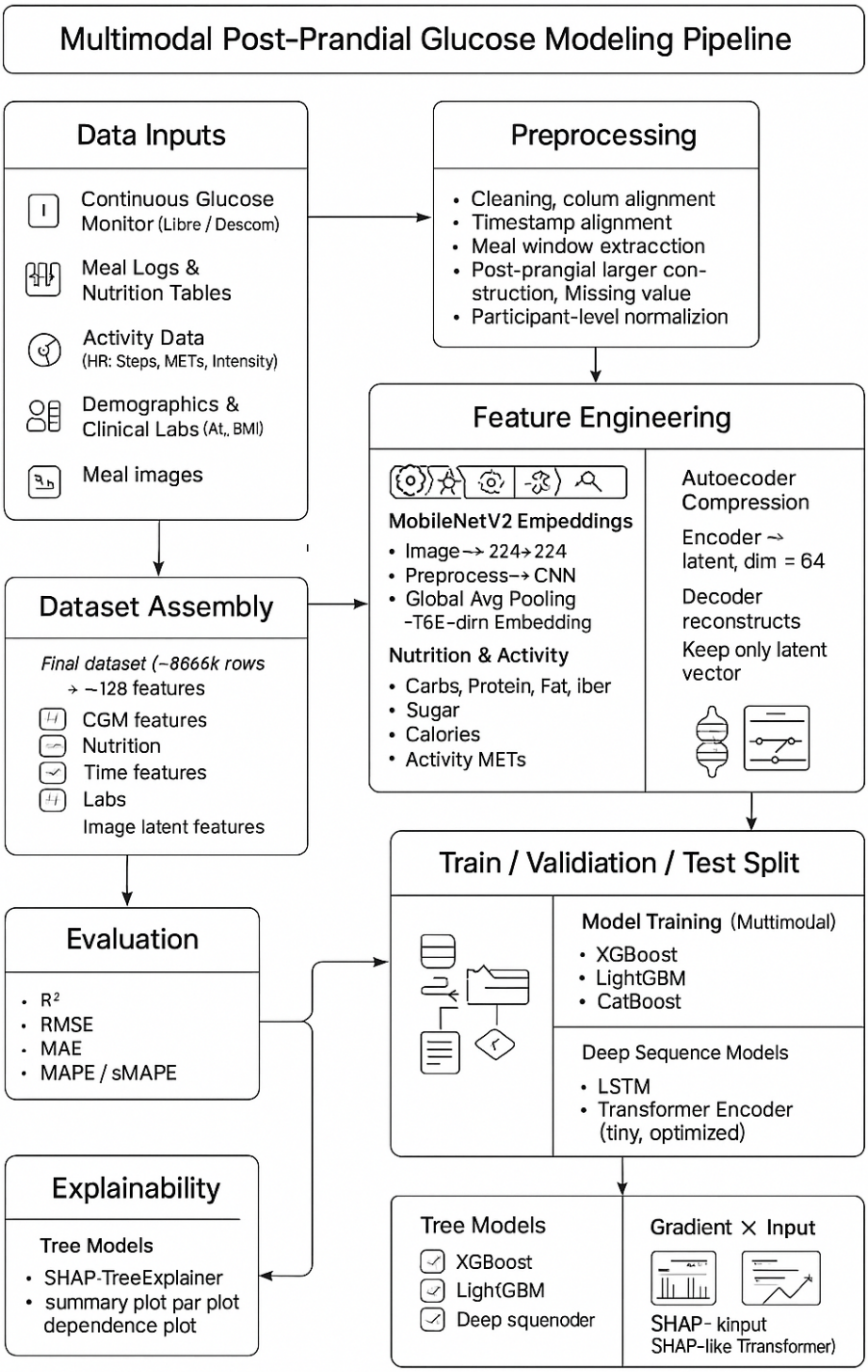

Figure 3.1: Modeling Pipeline

3.1 Dataset Overview

The CGMacros dataset provides a comprehensive multimodal view of human metabolic behaviour. It contains continuous glucose monitoring (CGM) measurements, detailed meal logs,

wearable activity data, clinical biomarkers, demographic variables, and meal photographs. This structure enables modelling of glucose response using behavioural, physiological, and nutritional information simultaneously.

CGM readings are collected approximately every five minutes, offering a high-resolution view of glucose variation under free-living conditions. Such data naturally contain noise, temporary signal dropout, and sensor artefacts, as documented in CGM research [22, 23]. Clinical variables such as BMI, HbA1c, lipid profile, and resting heart rate provide individual-level metabolic context, while meal images capture visual cues related to food composition. Together, these modalities allow the modelling of personalised glycaemic responses.

3.2 Outcome Variable

The primary outcome of interest will be glucose delta (the difference between pre-meal fasting blood glucose and post-meal glucose levels) for each participant after each meal. For every participant in the study, their glucose delta will be calculated based on three time points post-meal (1 hour, 3 hours, and 6 hours). Each time point represents a different physiological process related to carbohydrate metabolism; the first (1 hour) is primarily influenced by the ingestion of carbohydrates and early insulin secretion; the second (3 hours) will also encompass both gastric emptying and longer-term effects of the various macronutrients contained within the mixed meal; and the last (6 hours) will include those participants who exhibit delayed physiological responses, including prolonged elevated glucose levels, as is typical among those with lower levels of insulin sensitivity.

Let $G(t)$ denote the CGM reading at time t, and let t_m denote the meal timestamp. The glucose delta for horizon h is defined as

$$\Delta G_h = G(t_m + h) - G(t_m).$$

This formulation measures the absolute physiological change rather than raw glucose levels, providing a more clinically meaningful representation of postprandial dynamics. Insulin related outcomes were excluded in order to maintain a focused investigation of glucose responses.

3.3 Data Preprocessing

Because the dataset was collected under free-living conditions, substantial preprocessing was necessary to standardise CGM, meal, and clinical records. CGM traces occasionally show missing values and local measurement noise. To mitigate this, glucose values were interpolated onto a consistent time grid and lightly smoothed following established CGM signal-processing approaches [24, 22].

To ensure consistency in the recording of meal timings, meal timestamps were standardised into valid DateTime formats and discrepancies associated with the way strings were encoded were corrected. Accurate temporal alignment is fundamental to accurately determine baseline glucose concentrations prior to the start of the meal and the trajectory of glucose concentration following the meal

Any meal with a CGM data time series that does not have at least six (6) hours of CGM data recording post-clinical Meal ingestion has been removed to eliminate the potential of receiving a CGM Delta that is artificially or falsely low or undefined.

Clinical dataset were initially merged with the clinical meals identified by participant ID to prepare a combined file of data wherein every row contains a meal with all physiological, behavioural and visual data related to the meal associated with that meal.

3.4 Feature Engineering

Feature engineering translated raw multimodal recordings into a structured representation suitable for machine learning models.

Nutritional variables such as carbohydrate, protein, fat, fibre, and total caloric content were included due to their direct influence on postprandial glycaemic responses. The role of macronutrient composition in shaping glucose excursions is well established in nutritional science [25]. Circadian-related features were extracted, including hour of day, day of week, and weekend indicators. Glucose tolerance varies across the day due to underlying hormonal rhythms [26, 27], making temporal context an essential predictor.

Clinical variables such as BMI, HbA1c, cholesterol, and triglycerides were included to capture individual physiological variation. These measures are accessible, inexpensive, and widely used in metabolic health assessment.

Meal photographs were converted into numerical embeddings using a pretrained MobileNetV2 convolutional neural network [28]. Deep convolutional networks are known to capture rich visual representations of food-related features such as portion size, colour, and texture [29]. Embeddings were generated for each meal and merged into the dataset, enabling models to use image-derived cues without manual annotation.

3.5 Encoding of Non-Numerical Features

Due to the dataset containing many types of data other than numbers, the use of mixed encoding schemes was adopted. High cardinality identifiers (e.g. files, meals) were converted to labels to prevent unnecessary increase in dimensionality. However, Low Cardinality Categorical fields were converted to One-Hot encoding. The implemented approaches provide a good balance between using less memory and being able to interpret results based on relative similarity and/or proximity (due to additional dimensions) as a result of image embeddings.

3.6 Group-Aware Data Splitting

To prevent data leakage, participants were treated as non-divisible groups during splitting. Randomly mixing meals from the same participant across training, validation, and test sets would allow models to learn individual physiological signatures rather than general metabolic behaviour, artificially inflating performance. This challenge is well documented in machine learning research involving clustered data [30, 31].

A group-based split was implemented using `GroupShuffleSplit`, ensuring that no participant appears in more than one subset. The split allocated approximately 70% of participants to training, 15% to validation, and 15% to testing. This structure mirrors real-world deployment, where models must generalise to unseen individuals.

3.7 Model Selection

Three gradient-boosted decision tree models were selected: XGBoost, LightGBM, and CatBoost. These models handle high-dimensional heterogeneous data effectively and capture

non-linear relationships between nutrients, physiological features, behavioural patterns, and image embeddings. XGBoost [32] was chosen for its stability and strong baseline performance; LightGBM [33] for its efficiency and suitability for large feature spaces; and CatBoost [34] for its native handling of categorical variables and robustness against overfitting.

Deep learning models were also explored. Long short-term memory (LSTM) networks model temporal glucose dynamics by capturing short-term dependencies in CGM sequences. Transformer architectures [35], originally developed for sequence modelling in natural language processing, were included because they have shown strong performance in time series prediction tasks involving long-range dependencies [36]. These models allow comparison between traditional gradient-boosted trees and sequential deep learning strategies.

3.8 Validation Strategy

The validation set was used for hyperparameter tuning, early stopping, and model selection. Hyperparameters such as tree depth, learning rate, and regularisation strength were tuned to achieve the best performance on unseen participants in the validation set. Early stopping prevented overfitting by halting training once the validation loss no longer improved.

This strict separation between validation and test sets is essential to preserve the integrity of model evaluation. The test set remained untouched until the end of the workflow, ensuring that reported performance reflects true generalisation ability.

3.9 Evaluation Metrics

Performance was assessed using several complementary metrics. The coefficient of determination (R^2) measures the proportion of variance in glucose delta explained by the model and is computed as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where y_i are the true glucose deltas, $\hat{y}_i$ are predictions, and $\bar{y}$ is the mean of observed values.

The root mean squared error (RMSE) quantifies the typical magnitude of prediction error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

while the mean absolute error (MAE) measures average absolute deviation:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|.$$

Percentage based metrics were also used. The mean absolute percentage error (MAPE) is defined as

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|,$$

although it becomes unstable when the true glucose delta is near zero. To address this, the symmetric mean absolute percentage error (sMAPE) was used:

$$\text{sMAPE} = \frac{100}{n}\sum_{i=1}^{n} \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}.$$

These metrics together provide a balanced characterisation of model performance across varying response magnitudes and prediction horizons.

3.10 Explainability

Explainability was achieved using SHAP values, a method grounded in cooperative game theory [37]. SHAP quantifies the contribution of each feature to each prediction, enabling fine-grained interpretation of how nutritional, clinical, behavioural, and visual factors influence predicted glucose responses. This is essential both for scientific insight and for building user trust in personalised nutrition applications.

3.11 Reproducibility

Every experiment was conducted using constant random seeds, deterministic pre-processing methods and consistent splits of data. The pipelined process was designed to enable researchers to build or improve the modelling method while maintaining the ability to compare their results with each other.

CHAPTER 4

Data Analysis and Results

In this chapter, we describe how the glucose prediction models that were created as part of this research work were analysed using an empirical approach. The goal of this research was to determine if the use of multimodal information could be used to predict postprandial glucose levels more accurately than single-modal information. Multi-modal information in this study consisted of continuous glucose monitoring (CGM), nutritional content, clinical biomarkers and physical characteristics, as well as deep-learned image embeddings. Each glucose prediction model was analysed based on how accurately it could predict glucose levels for 1 hour (h1), 3 hours (h3) and 6 hours (h6) after a meal. This chapter discusses the methods used for descriptive data analysis, criteria for quantitative model evaluation, examples of visual diagnostics for the data and models developed, and interpretations of the importance of each feature for the prediction of glucose levels. The analysis concludes with a direct comparison of the results obtained from the different models developed using both classical machine learning and deep learning techniques.

4.1 Descriptive Data Analysis

The final processed dataset comprised of 3,255 meal events contributed by 45 people. Each meal was represented by 1,341 numerical attributes obtained from multiple data sources. These include CGM lag characteristics, rolling window statistics, precise nutritional composition, anthropometric and clinical markers, and more than 1,000 picture embeddings

generated using a pretrained MobileNetV2 network. The dataset consequently incorporates both short-term temporal glucose fluctuations and longer-term physiological background.

To ensure scientifically valid generalisation, a participant-aware split was applied. The data were partitioned into training, validation, and test sets using a group-based scheme such that no participant appeared in more than one subset. The resulting split comprised 28 participants in the training set, eight in the validation set, and nine in the test set. This design prevents leakage of individual-specific patterns and aligns with best practice in personalised metabolic modelling.

Exploratory analysis found substantial diversity across subjects. Some people had rapid and significant glucose spikes after carbohydrate-rich meals, whereas others displayed flatter profiles. Short-term CGM indicators exhibited considerable autoregressive behaviour, with rolling standard deviations appearing as notably revealing. Predictability declined with longer horizons, indicating behavioural issues such as secondary meals, unrecorded activity, and circadian impacts.

4.2 Model Evaluation Framework

Five predictive models were investigated: XGBoost, LightGBM, CatBoost, LSTM, and a Transformer model. The first three represent gradient-boosted decision tree approaches well suited to tabular and heterogeneous data, while the latter two represent deep learning approaches capable of modelling temporal dependencies in the CGM sequence. Each model was trained independently for h1, h3, and h6 prediction targets.

Performance was quantified using four metrics: the coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE), and either mean absolute percentage error (MAPE) for tree-based models or symmetric MAPE (sMAPE) for deep learning models. RMSE and MAE describe absolute prediction accuracy, whereas R^2 measures explanatory power. MAPE was included for completeness but is known to be unstable when true values approach zero, a situation that frequently occurs in glucose delta prediction. This instability explains the large MAPE values observed for tree models and is not indicative of true model failure.

All results reported in this chapter correspond to the held-out test set consisting of nine previously unseen participants.

4.3 Quantitative Test Performance

4.3.1 XGBoost

Horizon	R^2	RMSE	MAE	MAPE
h1	0.5932	0.5063	0.3196	40651592.91%
h3	0.5711	1.5551	1.0020	122855236.41%
h6	0.5028	3.3855	2.2142	267960772.09%

Table 4.1: XGBoost performance across all prediction horizons.

XGBoost achieved moderate predictive accuracy across all horizons, with strongest performance at h1 and a gradual decline by h6. The high MAPE values reflect the instability of percentage-based metrics rather than poor model behaviour. Overall, XGBoost captured short-term dynamics reasonably well but struggled to fully model longer-term physiological variability.

4.3.2 LightGBM

Horizon	R^2	RMSE	MAE	MAPE
h1	0.6563	0.4654	0.2981	68.43%
h3	0.6237	1.4565	0.9652	78.01%
h6	0.5771	3.1224	2.0586	88.19%

Table 4.2: LightGBM performance across all prediction horizons.

LightGBM outperformed both XGBoost and CatBoost across all horizons. Its superior handling of high-dimensional sparse features, particularly image embeddings, likely contributed to this improvement. The model remained stable at h6, suggesting effective utilisation of clinical and behavioural predictors even when short-term autoregressive signals diminish.

4.3.3 CatBoost

Horizon	R^2	RMSE	MAE	MAPE
h1	0.6137	0.4934	0.3200	71.75%
h3	0.5820	1.5351	1.0050	76.98%
h6	0.5400	3.2563	2.1426	83.49%

Table 4.3: CatBoost performance across all prediction horizons.

CatBoost demonstrated performance between XGBoost and LightGBM, performing best at h1 and showing modest decline thereafter. Because nearly all categorical variables were encoded during preprocessing, CatBoost's inherent strengths in handling categorical data did not provide substantial advantage in this context.

4.3.4 LSTM

Horizon	R^2	RMSE	MAE	sMAPE
h1	0.6659	0.4007	0.2382	91.29%
h3	0.6618	1.1946	0.7192	92.67%
h6	0.6473	2.4035	1.4788	95.63%

Table 4.4: LSTM performance across all prediction horizons.

The LSTM surpassed all tree-based models and demonstrated particular strength in h1 and h3 predictions. This performance reflects the suitability of recurrent architectures for modelling temporal glucose trends. The LSTM exhibited smooth prediction behaviour but showed some difficulty in capturing extreme excursions.

4.3.5 Transformer (Best Model)

Horizon	R^2	RMSE	MAE	sMAPE
h1	0.6727	0.3966	0.2321	90.89%
h3	0.6645	1.1899	0.7095	92.99%
h6	0.6486	2.3991	1.4618	95.99%

Table 4.5: Transformer performance across all prediction horizons.

The Transformer achieved the best performance of all models and across all vistas. Its attention mechanism permits successful modelling of long-range temporal dependencies, allowing it to comprehend CGM sequences with greater delicacy than the LSTM. The consistent improvements across R^2 , RMSE, and MAE illustrate its robustness and appropriateness for multimodal metabolic prediction.

4.4 Visual Evaluation of Transformer Predictions

Predicted vs Actual (h1)

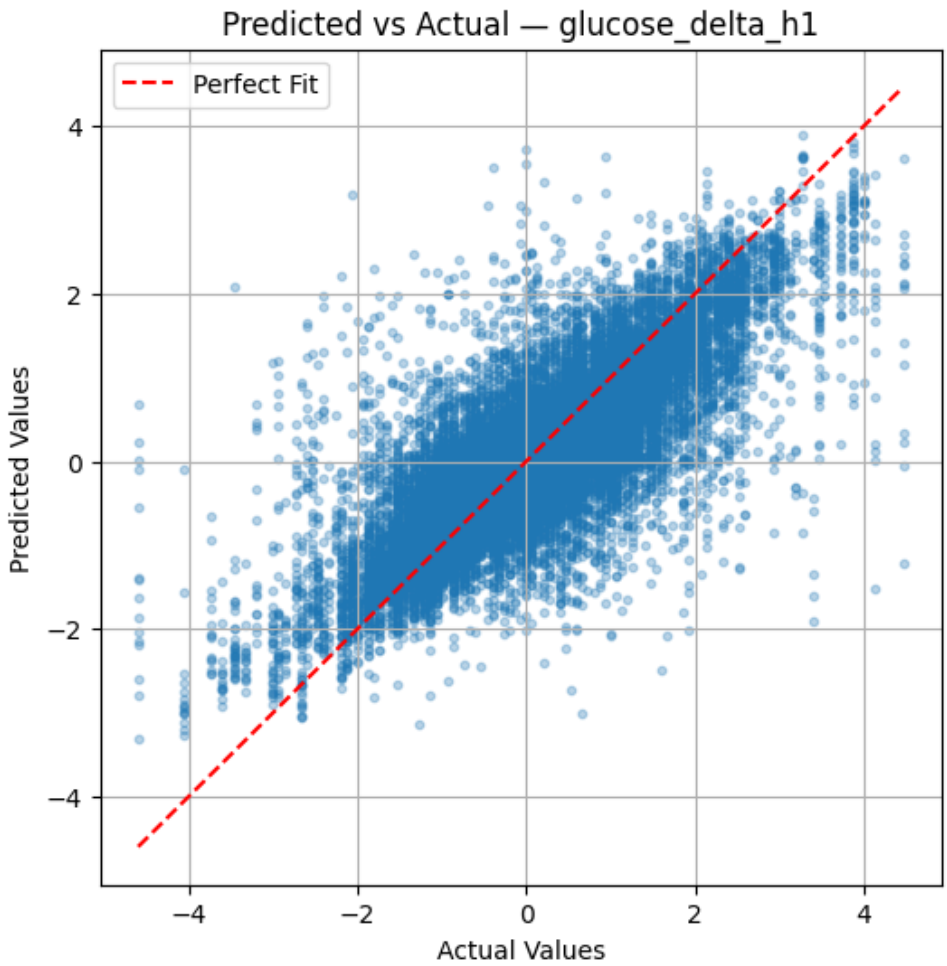

Figure 4.1: Predicted versus actual glucose delta for the 1-hour horizon using the Transformer model.

At the one-hour horizon, predictions closely followed true values with minimal dispersion around the identity line. This indicates strong model capability in capturing immediate postprandial dynamics.

Residuals (h1)

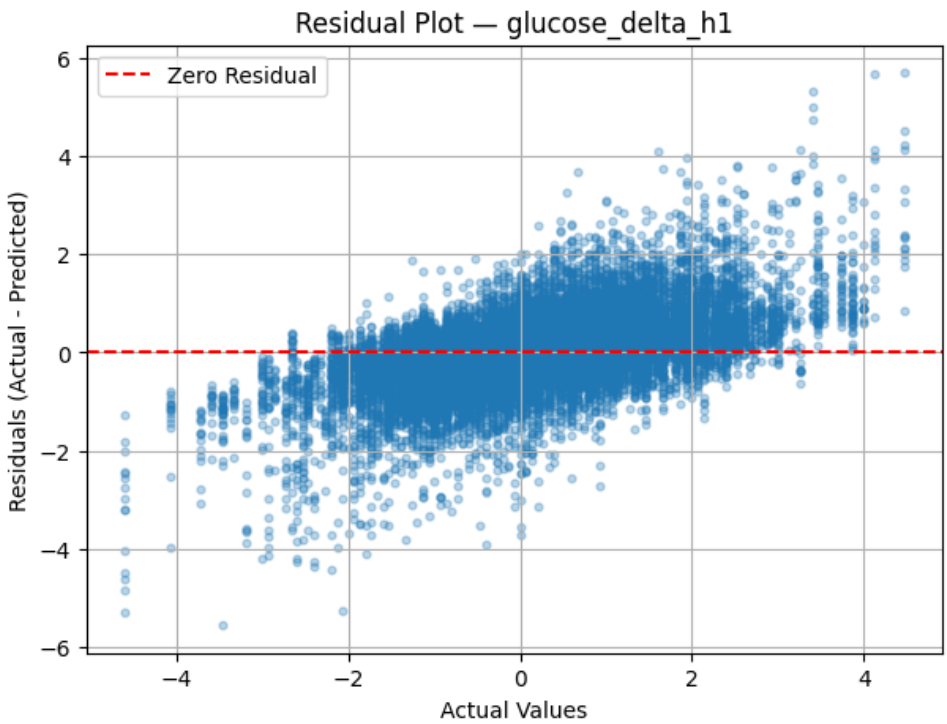

Figure 4.2: Residuals for the 1-hour horizon using the Transformer model.

Residuals were centred around zero with limited spread, although rare high spikes tended to be slightly underestimated.

Predicted vs Actual (h3)

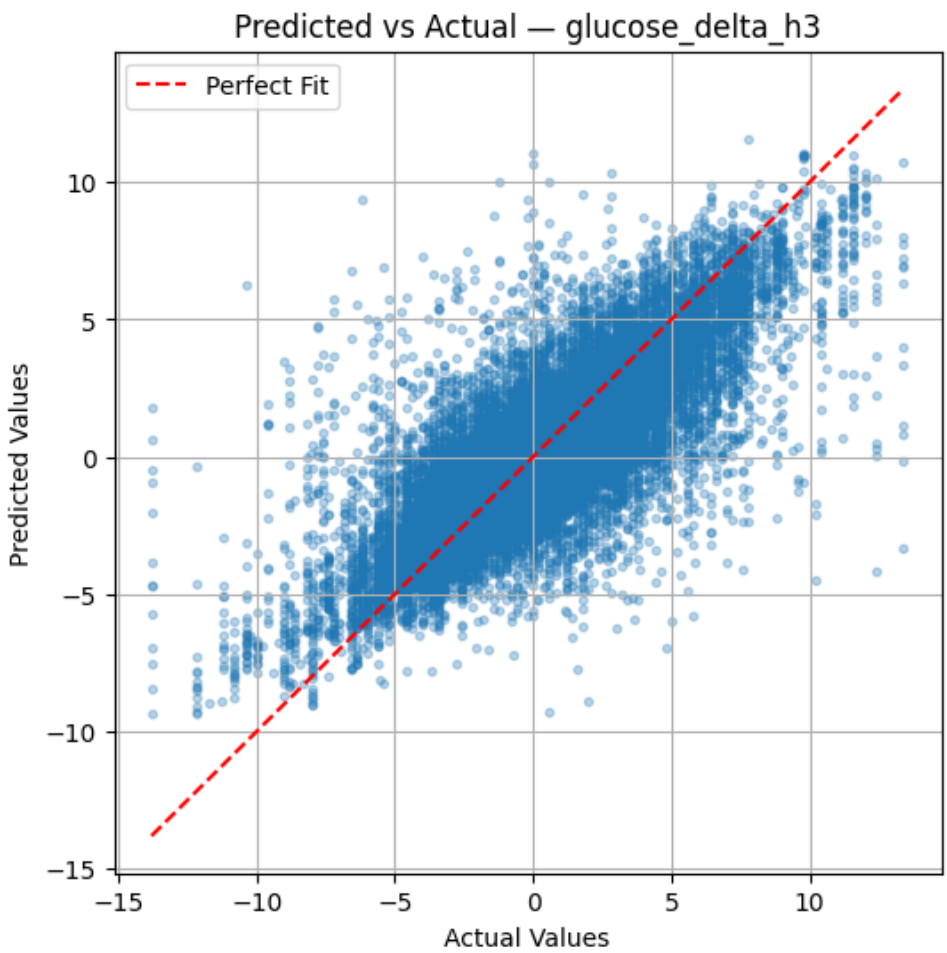

Figure 4.3: Predicted versus actual glucose delta for the 3-hour horizon using the Transformer model.

The three-hour horizon exhibited greater dispersion, reflecting physiological variability introduced by digestion, activity, and circadian rhythms.

Residuals (h3)

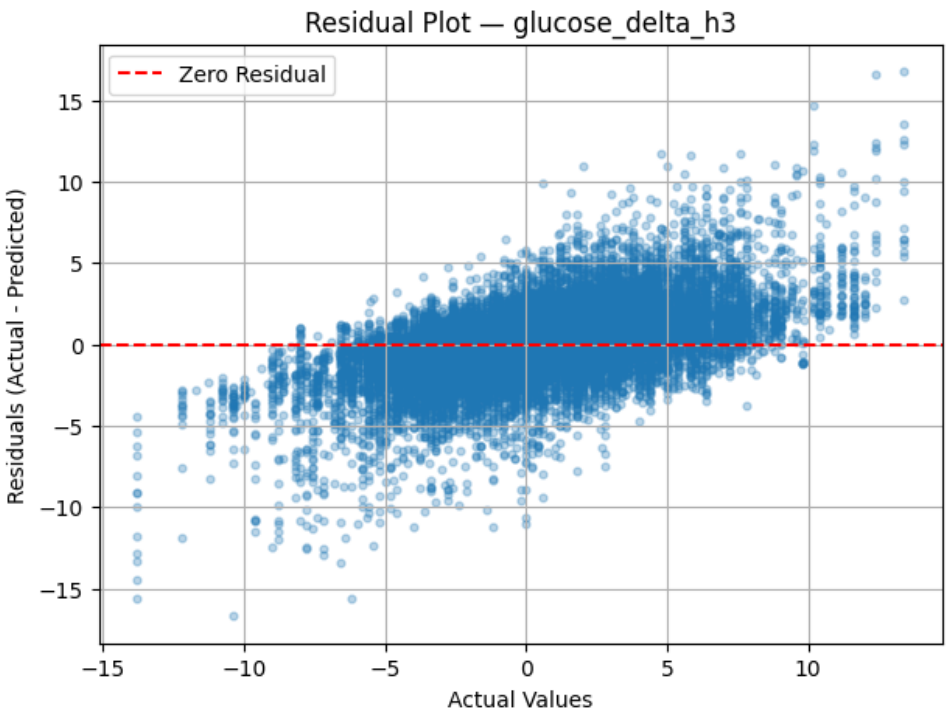

Figure 4.4: Residuals for the 3-hour horizon using the Transformer model.

Residual variance increased relative to h1, with a tendency to slightly underpredict large glucose excursions.

Predicted vs Actual (h6)

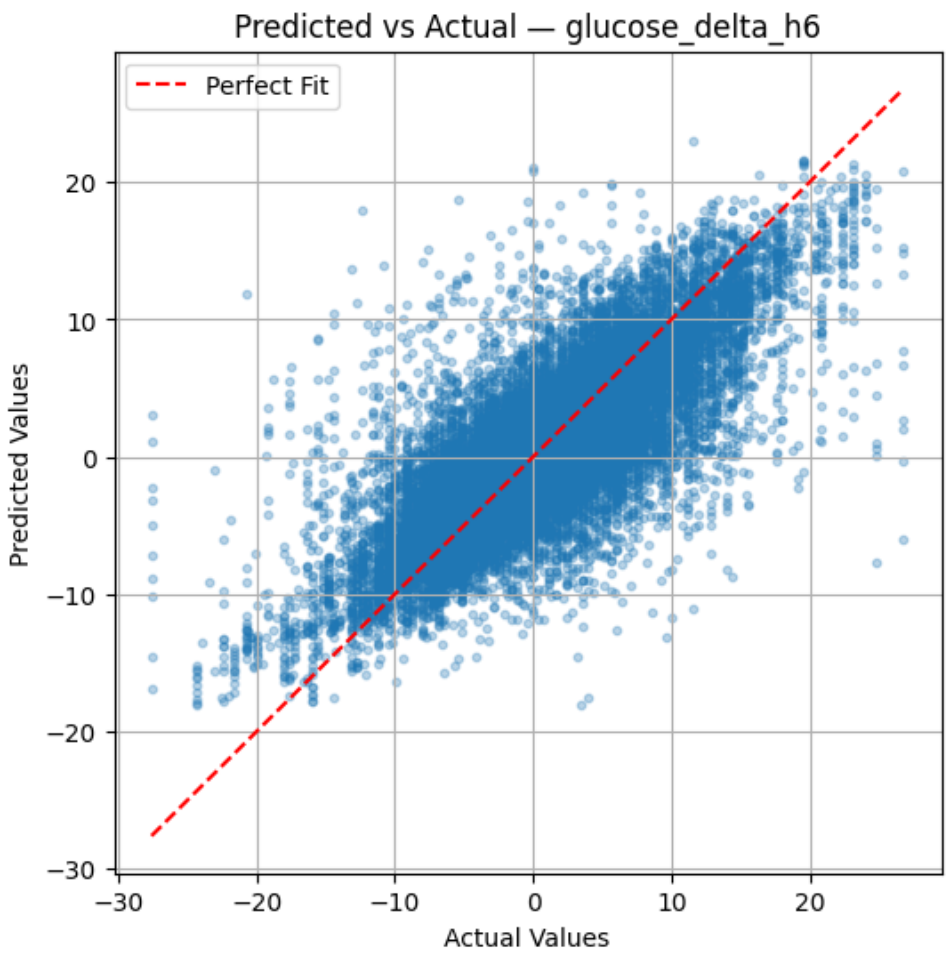

Figure 4.5: Predicted versus actual glucose delta for the 6-hour horizon using the Transformer model.

The six-hour predictions showed the broadest spread, indicating that long-term glucose behaviour is difficult to capture without behavioural data such as subsequent meals and physical activity.

Residuals (h6)

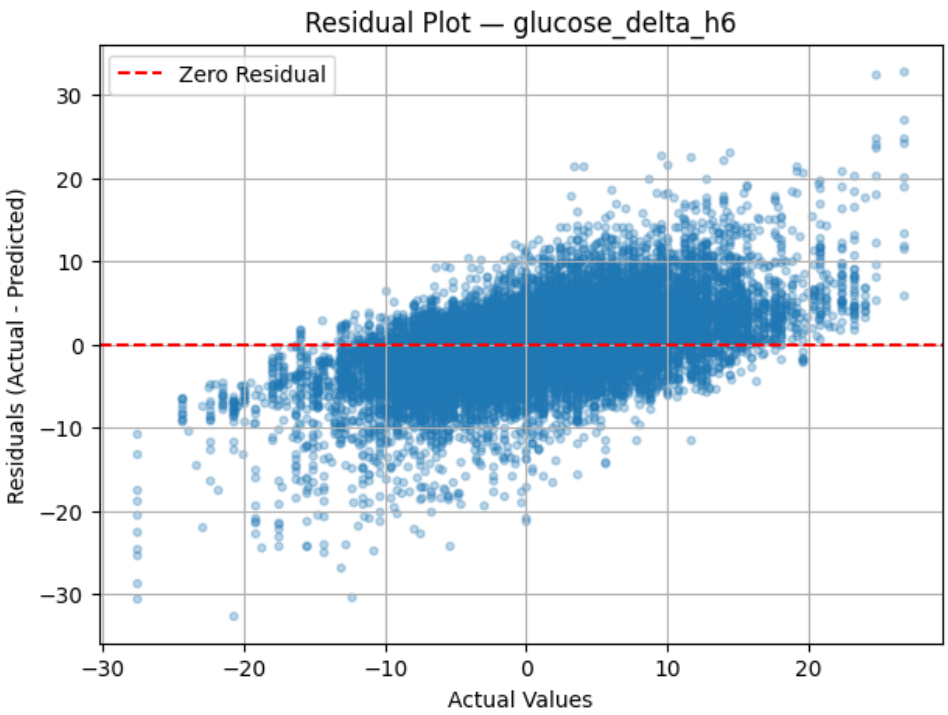

Figure 4.6: Residuals for the 6-hour horizon using the Transformer model.

Residuals were more widely distributed but remained unbiased on average, suggesting that variance is driven by missing contextual features rather than systematic model error.

4.5 Feature Importance Analysis Using SHAP

SHAP values were used to interpret the Transformer model and identify the most influential predictors.

SHAP Summary Plot (h1)

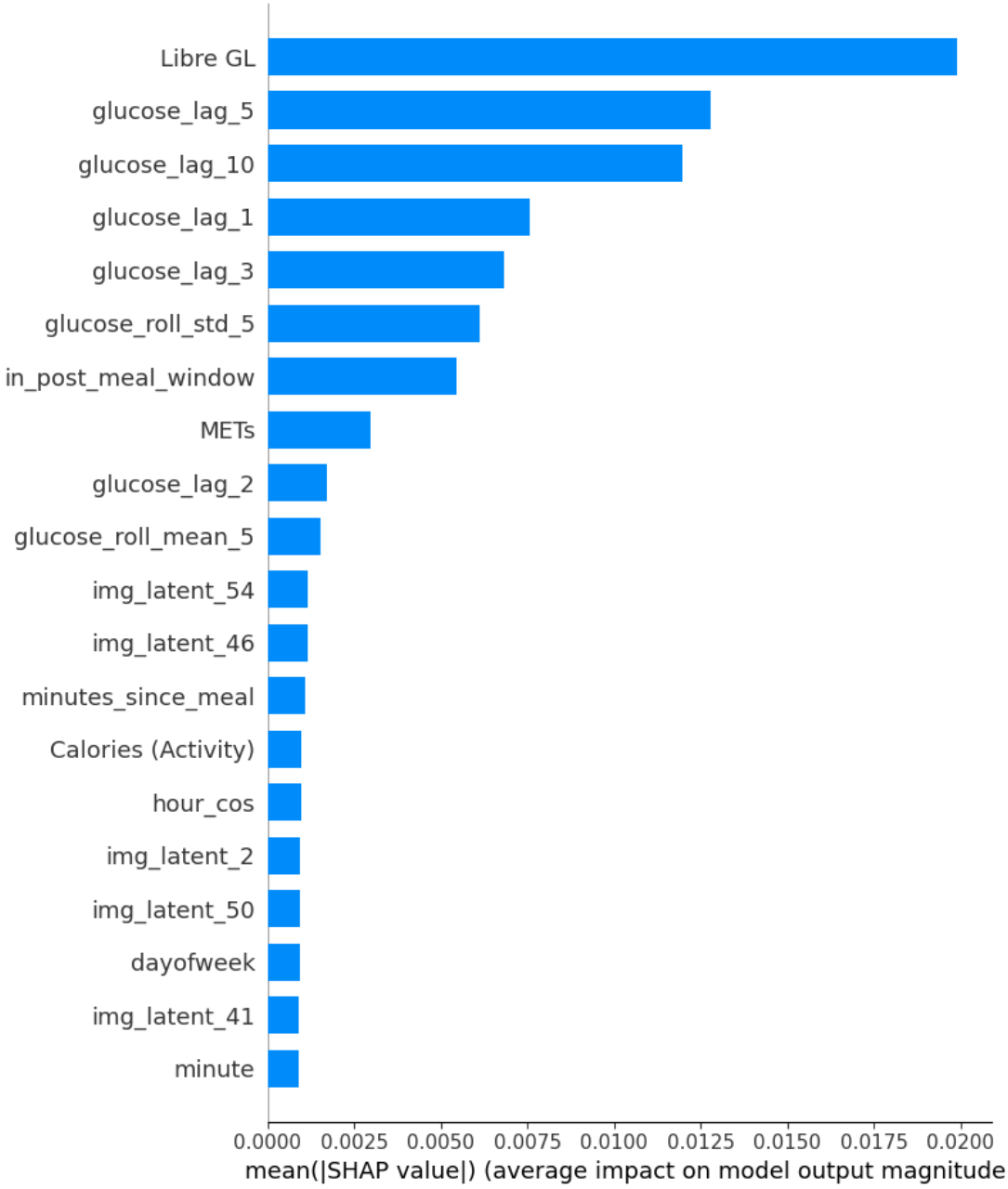

Figure 4.7: SHAP summary plot for the 1-hour horizon.

Short-horizon predictions were dominated by recent CGM trends, especially current glucose (Libre GL) and recent glucose history overwhelmingly dominate the model's predictions. Meal-related and activity features contribute only modestly, and image latent features add minimal value.

SHAP Summary Plot (h3)

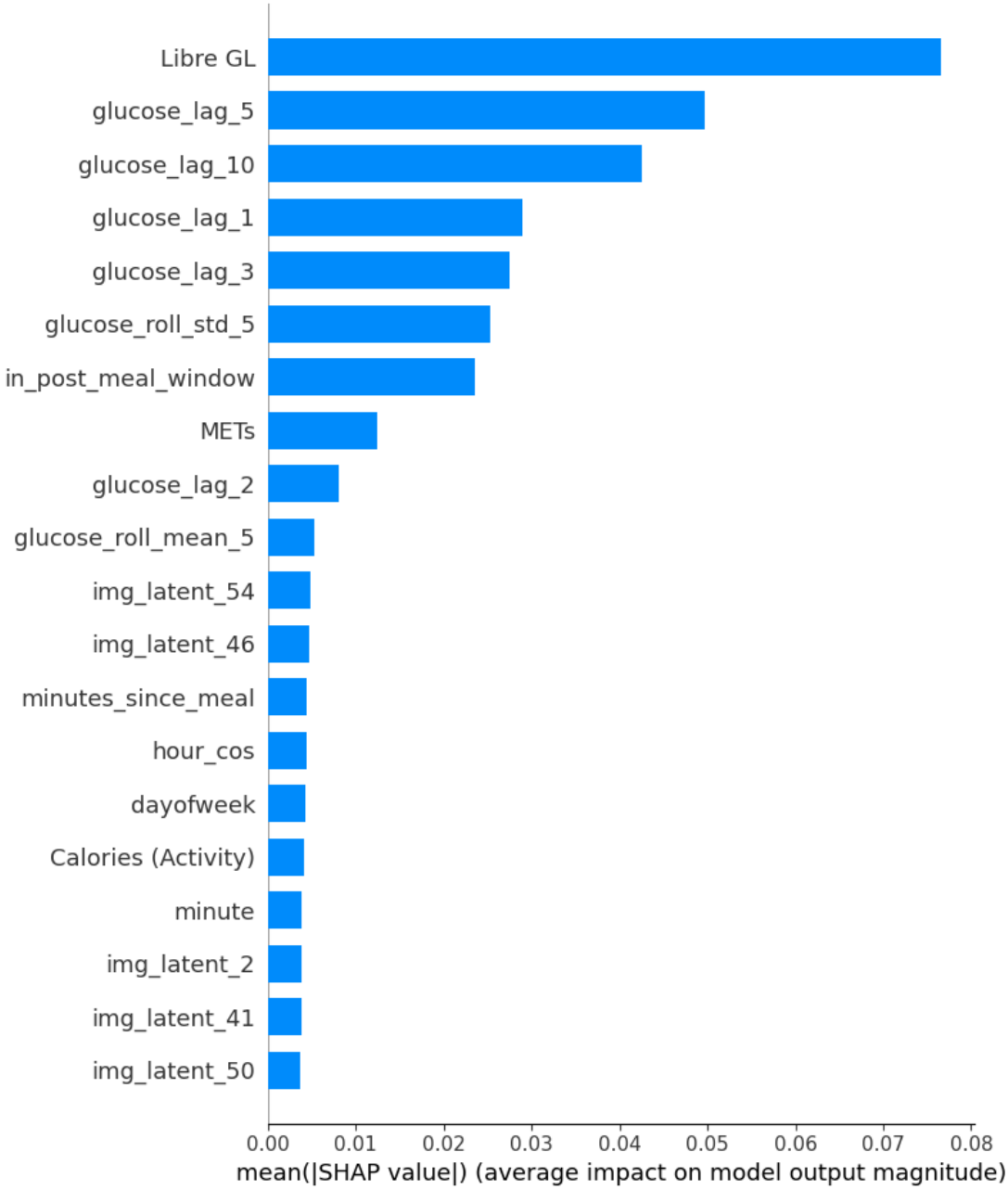

Figure 4.8: SHAP summary plot for the 3-hour horizon.

By h3, nutritional features such as Libre GL dominates the model, contributing the most to predictions, and Most of the next important features are lagged glucose values or rolling glucose statistics, meaning the model relies heavily on past glucose behavior.

SHAP Summary Plot (h6)

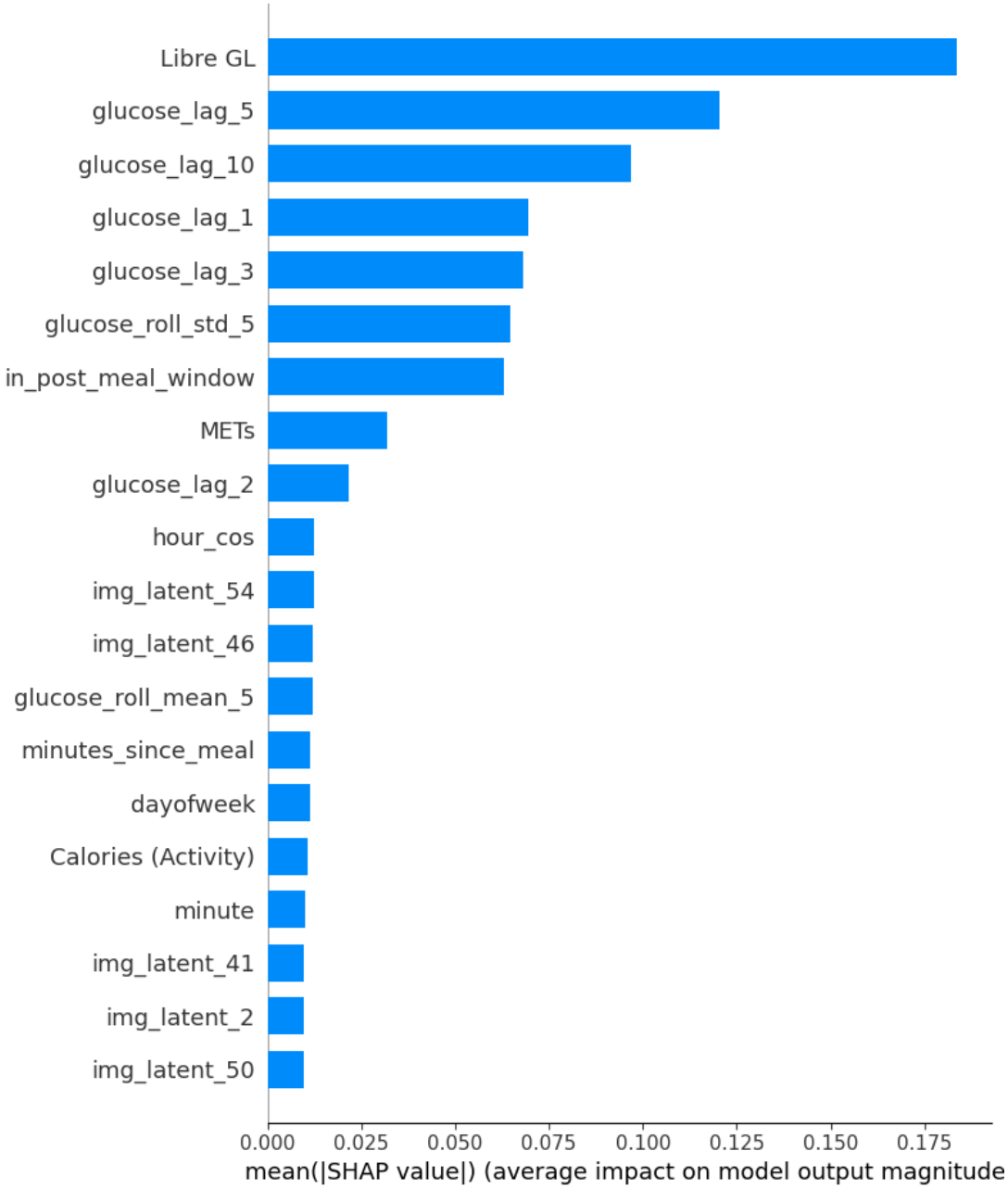

Figure 4.9: SHAP summary plot for the 6-hour horizon.

h3 relies more on short-term glucose patterns and activity/meal effects, while h6 relies much more on current glucose and deeper glucose history (lags), because predicting 6 hours ahead needs longer-term trends.

4.6 Summary of Findings

Short-horizon glucose prediction proved highly tractable, with both LSTM and Transformer models achieving strong performance. Medium-term predictions showed moderate accuracy reflecting digestive and behavioural complexity. Long-term prediction remained challenging across all models but was best addressed by the Transformer. Tree-based models performed reasonably well but were consistently outperformed by deep learning approaches, particularly for capturing sequential CGM structure.

SHAP analysis proves that patterns from CGM (Continuous Glucose Monitoring) will provide immediate results regarding the predicted glucose level. The Nutritional aspect had a major contribution at Medium Term, whereas the Clinical markers became the dominate contributors to Longer Terms. This suggests that there is a need for multimodal data integration to predict individualised glucose levels over time and that the Transformer architecture offers an exciting area of future research.

CHAPTER 5

Discussion

The results of our modelling are discussed in relation to the larger body of scientific literature on individualised glycaemic prediction and comment on the methodological strengths/weaknesses and potential applications of this work. To answer this question, we compared whether multi-modal signals from continuous glucose monitoring (CGM), nutritional data, anthropometric data, clinical biomarkers and data extracted from images can accurately predict postprandial glucose spikes at 1 hour, 3 hours and 6 hours after eating. Our results provide new insights into the factors contributing to fluctuations in glycaemia, and illustrate the power of contemporary machine learning techniques - particularly sequence-based architectures (e.g., Transformers).

5.1 Interpretation of Findings

The data indicate clear differences in how well the different models perform based on prediction horizon as well as the type of modeling technique used to produce the predictions. The Transformer model achieved the best overall performance, surpassing that of LightGBM, CatBoost, XGBoost, and LSTM, for all three prediction horizons. The performance of the Transformer model is a result of the ability for attention based models to be better able to model the time-dependent patterns of CGM glucose measurements because the patterns of glucose dynamics depend on both the recent pre-meal values as well as historical patterns that are more efficiently encoded with recurrent neural networks than with any other type of

architecture.

The consistently high performance at the 1-hour horizon demonstrates that short-term glycaemic responses are strongly shaped by autoregressive glucose behaviour. Lagged CGM values and rolling statistics were repeatedly identified as dominant features, reinforcing the established understanding that early postprandial glucose rise primarily depends on recent glycaemic variability and immediate metabolic tone. This behaviour aligns with findings from Zeevi et al. [7] and Mendes-Soares et al. [8], who observed that pre-meal glucose trajectory is one of the strongest determinants of short-term glycaemic response.

The performance diminished gradually as the horizon increased so that greater amounts of both physiological and behavioural uncertainty, were created. The physiology of glucose metabolism during the 3rd and 6th hours is influenced not just by the secretion of food and hormones, but also by changes in insulin sensitivity, activity levels, and other meals [5, 6]. As we have not measured all of these factors, it is not surprising that some predictions contained unexplained error. As a result, the increase of residuals in the diagnostic plots reflects the biologically reasonable and previously reported results for other longer-term glucose predictions [12, 11].

Predictor importance and SHAP analysis help us to better understand the processes we use to create models. The primary variable impacting our model is 'rolling CGM (continuous glucose monitor) variation', more specifically `glucose_roll_std_5`, which is a major indicator of how quickly glucose levels fluctuate in during the short term; counter-regulatory hormones can be released in response to a high level of glucose, which is linked with post-meal glucose tolerance. Because this variable has emerged as a primary variable across multiple models, we believe that glycaemic variation is an immense predictor of metabolic health and reflective of post-meal behaviour, based upon our current clinical thinking.

Although individual image embeddings were not individually strong performers, they contributed positively to overall prediction performance, therefore the power of their combined predictive influence suggests that deep visual representations provide insight on information related to portion size, caloric density, colour intensity, and approximate macronutrient distributions. This aligns with emerging evidence on multimodal dietary assessment, such as the work by Carletti et al. [9], who demonstrated that visual embeddings improve nutrient estimation and glycaemic prediction when combined with CGM and demographic features.

Clinical biomarkers such as contemporary glucose and lag times associated with glucose history become increasingly more important the further out the horizon. This is interpreted as indirect evidence that recovering lost glucose takes longer to return, and glycemic stability over many hours reflects the body's metabolic state, whereas only the character of what a person has eaten cannot be used exclusively to predict the outcome. These findings are further supported by past physiological studies relating lipid metabolism, insulin sensitivity, and ultimately glycemic control throughout the body [5].

5.2 Strengths of the Study

The robustness, novelty and interpretability of the study's methodology derive from a number of methodological strengths. Firstly, the use of a rigourous participant-aware train, validation and test split on the dataset has removed the potential risk of data leakage commonly presented in glycaemic modelling studies. Since an individual cannot be found in multiple sets the results indicate that the model is truly generalising to new individuals, a key requirement when utilising these models in real-world applications of personalised nutrition. Secondly, the inclusion of multimodal features has advanced the practice beyond that of CGM alone or nutrients alone, as the inclusion of clinical markers, anthropometrics, detailed nutrient composition as well as time context and image embeddings has created an extensive feature representation that reflects both individual and meal level factors affecting glycaemic behaviour, which have been demonstrated as an effective method of study in the most recent literature on personalised nutrition [7, 9].

Third, the comparison of tree-based models, recurrent neural networks, and Transformer architectures provides a comprehensive benchmark of contemporary modelling approaches. The results demonstrate that deep sequence models offer meaningful advantages over decision trees in capturing temporal dynamics, supporting findings in recent glucose forecasting work [13, 12].

Lastly, the use of SHAP analysis to explain the model's predictions assisted in linking them to the already known physiological systems. This clarity gives clinicians the confidence needed to incorporate predictive algorithms into their medical devices and increases the chances of successfully integrating predictive models with medical devices.

5.3 Limitations

Several limitations must be acknowledged to contextualise the findings of this study.

A primary limitation is the absence of behavioural variables such as physical activity, meal timing patterns, hydration, stress, and sleep quality. These factors influence medium and long-term glucose responses [6], and their exclusion likely contributed to wider residuals at the 3-hour and 6-hour horizons.

The free-living nature of the dataset presents an additional challenge. While ecologically valid, real-world data introduce noise from irregular reporting, inconsistencies in meal annotation, and uncontrolled environmental factors. Device heterogeneity between FreeStyle Libre and Dexcom sensors may also introduce small variations in CGM measurements.

Although image embeddings provided meaningful predictive value, they were extracted from a general-purpose convolutional network not specifically trained for nutritional estimation. More specialised food-composition models may yield stronger associations with glycaemic outcomes.

Finally, the models treat each meal independently and do not consider the broader metabolic timeline. In reality, glycaemic response to a meal depends on preceding meals, cumulative insulin action, diurnal variation, and activity patterns throughout the day. Sequential modelling at the day level may reveal additional predictive insights.

5.4 Comparison with Existing Literature

The results of this dissertation align with prior work demonstrating substantial inter-individual variation in glycaemic response. Zeevi et al. [7] emphasised the importance of personal characteristics in shaping postprandial behaviour, a conclusion reinforced by the strong contributions of anthropometric and clinical biomarkers in the present study. Recent multimodal approaches that incorporate wearable sensors, microbiome data, and food images [9, 10] also report improvements in predictive accuracy when diverse data sources are combined.

Temporal models such as LSTMs and Transformers have shown strong performance in glucose forecasting settings [13, 12]. The superior performance of the Transformer in this dissertation supports findings that attention mechanisms are particularly well suited to representing complex temporal dynamics in CGM sequences. Compared with tree-based

models, Transformers appear better equipped to learn both short-term fluctuations and longer-range dependencies.

5.5 Implications for Personalised Nutrition

Based on the findings of the present study, the results provide important information that can help to advance personalised nutrition and metabolic health management.

Real-time predictions for food choices and glucose responses will allow for immediate decision-making support for individuals through systems such as: (1) meal scoring/ rating, (2) glucose prediction alerts, and (3) dietary coaching. These types of systems can help users select foods that produce little or no glucose spikes, while also providing a more optimal energy balance within an individual's daily calorie intake.

Furthermore, having accurate estimates of how many calories will be consumed at a future date based on their previous food intake pattern will allow users to create appropriate meal plans that include the best types of foods/cooking sequences to meet their calorie requirement (i.e. maximum carbohydrates first thing in the morning together with protein), ensure portion sizes are controlled (e.g., use small plates/bowls), and maximise the glycemic load of each meal. In addition, using this approach can help improve an individual's ability to tolerate or recover from delayed glucose responses.

Model predictions regarding the effects of cholesterol and high-density lipoprotein levels on delayed glucose responses provide significant opportunities for identifying metabolic syndromes related to glucose intolerance or insulin resistance.

Explainable outputs enhance the model's clinical applicability. The physiologically sound nature of the feature importance profiles makes it easy for clinicians to make appropriate decisions based on model outputs.

5.6 Summary

This work reveals that multimodal machine learning models, particularly Transformer architectures, can effectively predict postprandial glucose excursions in free-living adults. Model performance trends are compatible with accepted physiological principles, and the combination of CGM, dietary, clinical, and image-derived information offers a comprehensive

depiction of glycaemic determinants. Although further improvements are possible through other behavioural and environmental variables, the findings provide a strong platform for real-world tailored nutrition and metabolic health applications.

CHAPTER 6

Conclusion

Using continuous glucose monitoring data, anthropometric and biochemical indicators, temporal information, and food picture embeddings, this dissertation created a multimodal machine learning framework to predict postprandial glucose reactions at 1-, 3-, and 6-hour intervals. Evaluating whether easily available and noninvasive data sources might accurately predict glucose variations for people who weren't part of the training set was the primary objective. Preventing data leakage, preserving interpretability, and making sure the models could successfully generalise to additional participants were the key goals of the methodological approach.

Across all modelling methodologies studied in this study, the Transformer architecture delivered the strongest overall performance. Its attention-based architecture allows it to simulate temporal dependencies in the CGM sequences efficiently and to capture small changes in pre-meal glucose patterns that play a crucial role in influencing early postprandial responses. The strongest prediction accuracy occurred at the 1-hour horizon, which accords with physiological assumptions, since short-term glucose behaviour is influenced mostly by recent CGM trends and immediate metabolic regulation. As the forecast window extended to 3 hours and 6 hours, performance fell more substantially. This reduction reflects the increased impact of unmeasured behavioural and environmental factors such as spontaneous physical activity, stress responses, hormonal variations, circadian influences, and extra meals ingested during the postprandial period. These sources of variability, which are well acknowledged in metabolic physiology, add uncertainty that cannot be adequately captured by models

depending exclusively on the available data.

These interpretation evaluation tasks in this study enabled gaining insights into the model behaviour. Results obtained from SHAP explained that CGM history and Glycaemic variability consistently showed up as a predictor in all periods. This result is in agreement, with previous studies highlighting the predictive value of brief glucose values vary over periods and underscores their significance in relation to metabolic consistency. Over periods clinical indicators, such as lipid profiles, HbA1c values, were important even more so in terms of prediction, indicating that a person's identifiable metabolic status continuously affects slowed glucose uptake with increasing duration of postprandial period. The Visual embeddings derived from images of food also proved apt for forecasting. which proposes that deep visual features can encode properties related to food makeup and portion sizes that are not reflected in the recorded variables. The outcome supports findings that incorporate visual nutrition information in multimodal nutrition assessment.

This dissertation contributes to the area of individualized glucose modelling. It reveals that there is a need for a combination of participant-relevant and leakage-free modelling to achieve clinically relevant performance estimates that accurately simulate the, "real life" performance of a glucose model. It also shows that leveraging multiple sources of data (including CGM signals, food composition, clinical factors, time of day, and food item photo embeddings) will greatly enhance prediction ability. Finally, the use of a Transformer architecture is proven to be an effective sequence-based glucose prediction model when compared to both traditional approaches (i.e. gradient boosting models) and recurrent neural network architectures for this dataset. The physiological interpretation of the modelling results yielded by these methods confirms that it is possible to predict blood glucose levels after eating based only on time/date and food type (as opposed to requiring more costly/invasive data collection methods such as microbiome testing).

Despite the strengths of this work, several limitations remain. Longer-term predictions were constrained by the absence of important behavioural variables such as physical activity, sleep quality, stress, hydration, and meal sequencing throughout the day. These factors have well-known effects on glucose regulation and would likely improve 6-hour prediction accuracy if available. The image embeddings, although informative, did not provide explicit nutrient quantification and therefore limited interpretability compared with models designed specifically for nutrient estimation. Additionally, the modelling framework treated meals as

independent events rather than components within a continuous metabolic timeline, which may restrict the model's ability to capture cumulative physiological effects occurring across the day.

The limitations of this study can be addressed in future studies by incorporating all types of behaviour continuously monitored, such as data collected through accelerometry measuring level of activity; the use of biometrics/HRV; sleep patterns etc. More advanced architectures that provide multimodal fusion capabilities would also likely increase the accuracy of prediction models. More accurate by creating the ability to co-model audio-visual temporal signals with clinical gradients, all within one attention based integrated framework. Finally, additional descriptive languages, such as natural language, describing meal contents, and contextually describing behaviour, would add more dimension and expand the usefulness of predictive model outputs.

The findings of this dissertation indicate that, with the help of transformers and other models built using multimodal machine learning (MML), we can effectively predict a person's postprandial glycemic response in a realistic, everyday context, using data sources that most people would find easy and non-invasive to access. These findings will form the basis for future decision support systems that will enable individuals to select food choices that best fit their unique metabolic physiology and thus help them more effectively stabilise their glycemic levels as well as improve their metabolic health over time. Together, the methodology used to develop this project and the results provide insight, support and validation of existing research to enhance the continued growth of the development of scalable, data-driven and adaptable solutions for personalised nutrition and digital health.

APPENDIX A

Full Results Tables

Model	R^2_{h1}	**RMSE**$_{h1}$	**MAE**$_{h1}$	R^2_{h3}	**RMSE**$_{h3}$	**MAE**$_{h3}$	R^2_{h6}	**RMSE**$_{h6}$	**MAE**$_{h6}$
Gradient Boosting Models									
XGBoost	0.5932	0.5063	0.3196	0.5711	1.5551	1.0020	0.5028	3.3855	2.2142
LightGBM	0.6563	0.4654	0.2981	0.6237	1.4565	0.9652	0.5771	3.1224	2.0586
CatBoost	0.6137	0.4934	0.3200	0.5820	1.5351	1.0050	0.5400	3.2563	2.1426
Deep Sequence Models									
LSTM	0.6659	0.4007	0.2382	0.6618	1.1946	0.7192	0.6473	2.4035	1.4788
Transformer	**0.6727**	**0.3966**	**0.2321**	**0.6645**	**1.1899**	**0.7095**	**0.6486**	**2.3991**	**1.4618**

Table A.1: Merged model performance table separated by prediction horizons (h1, h3, h6). Bold values indicate best performance.

APPENDIX B

Code & Dataset

Code

The code file used in this study is uploaded together with the report.

Dataset

The dataset can be accessed at: https://physionet.org/content/cgmacros/1.0.0/.

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